FAILURE OF ACC FOR MINIMAL LOG DISCREPANCIES OF KOLLÁR COMPONENTS

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Abstract. We show that the minimal log discrepancies of Kollár components of klt threefold singularities do not satisfy the ACC. Thus we answer negatively Question 7.3 in Zhuang's paper on boundedness of singularities and minimal log discrepancies of Kollár components, as well as the ACC part of Question 10.6.11 in his survey on stability of klt singularities.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

A set of real numbers satisfies the DCC (resp. the ACC) if it contains no infinite strictly decreasing (resp. increasing) sequence. A set of real numbers satisfying the DCC is called a *DCC set*. Shokurov's ACC conjecture for minimal log discrepancies [Sho88, Problem 5] predicts that for any positive integer d and any DCC set $\Gamma \subset [0,1]$, the set of $\mathrm{mld}(X \ni x, B)$, where $(X \ni x, B)$ runs over all lc pairs such that $\dim X = d$, x is a closed point, and $B \in \Gamma$, satisfies the ACC. See also [Sho04, p. 316, (ACC)]. Among the known cases are surfaces [Ale93, Theorems 3.2 and 3.8], toric germs ([Bor97, Corollary 2.1] when $B = 0$, and [Amb06, Theorem 1.1(2)] in general), germs admitting an ϵ-plt blow-up for a fixed $\epsilon > 0$ [HLS26, Theorem 1.1], and germs whose local volumes are bounded from below by a fixed positive number [HLQ24, Theorem 1.3]. We refer the reader to [HLS26, Definition 3.6] for ϵ-plt blow-ups. In dimension three, the ACC holds for closed points at which X is terminal and $\mathrm{mld}(X \ni x, B) \geq 1$ [HLL25, Theorem 4.1], for the values at least 1 with coefficients in a DCC set, for the standard coefficient set, and for quotients of smooth threefold germs by finite abelian groups [Liu26a, Theorems 1.2, 1.3, and 1.5].

Zhuang [Zhu24a] introduced the minimal log discrepancy of Kollár components in order to study klt singularities whose local volumes [Zhu24a, Definitions 2.7 and 2.8] are bounded below, see [HLQ23, Conjectures 1.6 and 8.9]. Let $(X \ni x, B)$ be a klt pair, where x is a closed point and $\dim X \geq 2$. A Kollár component over $(X \ni x, B)$ is a prime divisor E over X with $\mathrm{center}_X E = x$ which is extracted by a projective birational morphism $f : Y \to X$ such that $-E$ is $\mathbb{Q}$-Cartier and f-ample and $(Y, f_*^{-1}B + E)$ is plt near E. Kollár components over $(X \ni x, B)$ always exist ([Xu14, Lemma 1] when B is a $\mathbb{Q}$-divisor, and [HLS26, Lemma 3.7] in general). The minimal log discrepancy of Kollár components $\mathrm{mld}^{\mathrm{K}}(X \ni x, B)$ is the minimum of the log discrepancies $a(E, X, B)$, where E runs over all Kollár components over $(X \ni x, B)$ [Zhu24a, Definition 1.4].

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We refer the reader to Definition 2.3 for the precise definitions. Zhuang also defines $\mathrm{mld}^{\mathrm{K}}$ at points of X which are not necessarily closed [Zhu24a, Definition 1.4].

Zhuang conjectured that $\mathrm{mld}^{\mathrm{K}}$ is bounded from above by a constant depending only on the dimension, a positive lower bound of the local volume, and a finite set of rational numbers in $[0,1]$ containing the coefficients of the boundary [Zhu24a, Conjecture 1.7]. Zhuang proved Conjecture 1.7 of [Zhu24a] in dimension three [Zhu24a, Corollary 6.11], and Xu and Zhuang proved it at closed points in any dimension [XZ24, Theorem 1.3], and Han, Liu, and Qi proved the bound at closed points for boundaries with coefficients in any DCC set [HLQ24, Theorem 1.5]. See also [Zhu24a, Theorem 1.5] and [Zhu24b, Theorem 1.2] for how upper bounds of $\mathrm{mld}^{\mathrm{K}}$ are used to study klt singularities whose local volumes are bounded below. Without any assumption on local volumes, Zhuang conjectured that for any positive integer n and any DCC set $\Gamma \subset [0,1]$, the values of $\mathrm{mld}^{\mathrm{K}}$ of klt pairs (X,B) with $B \in \Gamma$ at points of codimension n are bounded from above by a constant depending only on n and Γ [Zhu24a, Conjecture 1.8]. Zhuang proved the bound at codimension two points when Γ is a closed DCC subset of $[0,1] \cap \mathbb{Q}$ [Zhu24a, Proposition 6.4]. Conjecture 1.8 of [Zhu24a] is proved in [Liu26b, Theorem 1.2].

Zhuang also showed that $\mathrm{mld}^{\mathrm{K}}$ behaves differently from mld. He constructed smooth surface germs with boundaries, each of which has only one Kollár component, such that the coefficients of the boundaries tend to 0 and the values of $\mathrm{mld}^{\mathrm{K}}$ tend to infinity [Zhu24a, Example 7.1]. He also showed that $\mathrm{mld}^{\mathrm{K}}$ is not lower semicontinuous [Zhu24a, Example 7.1]. In analogy with Shokurov's ACC conjecture for minimal log discrepancies, Zhuang asked the following question.

Question 1.1 ([Zhu24a, Question 7.3])**.** Let d be a positive integer and $\Gamma \subset [0,1]$ a DCC set. Does the set of $\mathrm{mld}^{\mathrm{K}}(X \ni x, B)$, where $(X \ni x, B)$ runs over all klt pairs such that $\dim X = d$, x is a closed point, and $B \in \Gamma$, satisfy the ACC?

In [Zhu24a, Question 7.3], B is a $\mathbb{Q}$-divisor. Question 1.1 allows $\mathbb{R}$-divisors.

In his survey, Zhuang asks whether the set of $\mathrm{mld}^{\mathrm{K}}(X \ni x)$ of klt germs $X \ni x$ of a fixed dimension satisfies the ACC, and whether it is bounded from above [Zhu25, Question 10.6.11]. The second part has a positive answer: by [Liu26b, Theorem 1.2] with $B = 0$ and $\Gamma = \{0\}$, $\mathrm{mld}^{\mathrm{K}}(X \ni x)$ is bounded from above in terms of $\dim X$.

When $d \leq 2$ and $B = 0$, the answer to Question 1.1 is positive. The case $d = 1$ is trivial. When $d = 2$, $\mathrm{mld}^{\mathrm{K}}(X \ni x) = \mathrm{mld}(X \ni x)$ for any klt surface germ $X \ni x$, since some divisor computing $\mathrm{mld}(X \ni x)$ is a Kollár component over $X \ni x$ [LX23, Theorem 1.2], see also [Zhu24a, Section 7]. Thus the answer is positive by the ACC for minimal log discrepancies of surfaces [Ale93, Theorems 3.2 and 3.8]. The reduction to mld is no longer available from dimension three on, even when $B = 0$ [Zhu24a, Example 7.2]. Thus, from dimension three on, we have to work with Kollár components directly.

In this paper, we answer Question 1.1 negatively: the ACC fails for $\mathrm{mld}^{\mathrm{K}}$ of klt germs of dimension three.

Theorem 1.2. *For any integer $m \geq 2$, let $r := 4m-1$ and $N := 2r+3$. Let X'_r be the hypersurface $(t^2 = y(x^2 + y^{2r} + y^3 z^N))$ in $\mathbb{A}^4_{x,y,z,t}$, let the group $\mu_r \subset \mathbb{C}^*$ of r-th roots of unity act on X'_r by $\zeta \cdot (x,y,z,t) = (x, \zeta y, \zeta^{-1} z, \zeta^{2m} t)$, let $X_r := X'_r/\mu_r$, and let $0 \in X_r$ be the image of the origin. Then X'_r is normal, K_{X_r} is $\mathbb{Q}$-Cartier, $X_r \ni 0$ is a klt germ of dimension 3, and*

$$\mathrm{mld}^{\mathrm{K}}(X_r \ni 0) = \frac{3m-1}{4m-1}.$$

In particular, the sequence $\{\mathrm{mld}^{\mathrm{K}}(X_{4m-1} \ni 0)\}_{m \geq 2}$ is strictly increasing.

Since $B = 0$ is a $\mathbb{Q}$-divisor and satisfies $B \in \Gamma$ for any Γ, the strictly increasing sequence in Theorem 1.2 shows that Question 1.1 and [Zhu24a, Question 7.3] have negative answers when $d = 3$ for any DCC set Γ, and that the ACC part of [Zhu25, Question 10.6.11] has a negative answer in dimension three. Together with the affirmative answer to Question 1.1 when $d \leq 2$ and $B = 0$, it follows that three is the minimal dimension of counterexamples to Question 1.1

with $B = 0$. Since $4(3m-1) < 3(4m-1)$, all the values $\mathrm{mld}^{\mathrm{K}}(X_r \ni 0)$ are less than $3/4$, in accordance with Conjecture 1.8 of [Zhu24a]. Forgetting t induces a double cover from X_r to the quotient of $\mathbb{A}^3_{x,y,z}$ by μ_r acting by $(x, \zeta y, \zeta^{-1} z)$, branched along the images of $(y = 0)$ and $(x^2 + y^{2r} + y^3 z^N = 0)$, and K_{X_r} is the pullback of the sum of the canonical divisor of the quotient and half of each image (Remark 3.4).

Sketch of the proof. By Lemma 3.3, $X_r \ni 0$ is klt and $X_r \cong Y_r/G_r$, where G_r is cyclic of order $2r$ (Construction 3.1 and Lemma 3.3(2)) and Y_r is the hypersurface

$$Y_r = (ab = u^{4r} + u^6 z^N) \subset \mathbb{A}^4.$$

By Definition-Proposition 2.6, a Kollár component E over $X_r \ni 0$ corresponds to a G_r-invariant Kollár component F over $Y_r \ni 0$ with $a(F, Y_r) = e\,a(E, X_r)$, where e is the number of elements of G_r acting trivially on F. Theorem 1.2 follows from four steps.

(i) Lemma 3.2(3) and the divisibility by e of the orders of the G_r-invariant functions z^r and $u^2 z$ along F prove Proposition 4.1 except when $\mathrm{mult}_F \mathrm{div}(u^2 z) = e$ and $\mathrm{mult}_F \mathrm{div}(z^r) \le \frac{r-3}{2} e$, that is, except for $2m - 2$ values of $\mathrm{mult}_F \mathrm{div}(z^r)/e$.

(ii) In each remaining case, e is even. Thus the involution $\tau \in G_r$ acts trivially on F, and a and b have proportional images in the graded algebra $\mathrm{gr}_F R$ (Definition-Lemma 2.5).

(iii) Since $\mathrm{gr}_F R$ is normal and N is odd, the equation of Y_r makes the image of z the square of a homogeneous element of degree smaller than the order along F of any function vanishing at 0, a contradiction.

(iv) The weighted blow-up of Y_r with weights $(2mr, 2mr, m, 2m-1)$ extracts a Kollár component F_r [Zhu24a, Lemma 2.5]. Since $e = r$ and equality holds in Lemma 3.2(3) for F_r, the induced E_r attains the bound (Construction-Proposition 5.1).

Items (i)–(iii) bound all Kollár components without determining them. Only Construction 3.1 and items (ii) and (iii) are new. Items (ii) and (iii) are **Steps 3 and 4** in the proof of Proposition 4.1.

(a) Passing to the cover Y_r. Since E in Proposition 4.1 is not assumed to be induced by a weighted blow-up or to be invariant under a torus action, $a(E, X_r)$ cannot be computed combinatorially. Instead, we bound log discrepancies of arbitrary prime divisors by orders of functions on Y_r, which is the double cover of X'_r adjoining a square root u of y (Lemma 3.3(3)). Indeed, expanding the generator $a^{-1} da \wedge du \wedge dz$ of ω_{Y_r} (Lemma 3.2(2)) at a general point of any prime divisor over Y_r implies Lemma 3.2(3). Since the quotient morphism $Y_r \to X_r$ is étale in codimension one (Lemma 3.3(4)), the bound passes to X_r through Definition-Proposition 2.6 (cf. [LX20, Lemma 2.13]). The exponents $4r$ and $N = 2r + 3$ in the equation of Y_r ensure that

$$\mathrm{mult}_F \mathrm{div}(u^6 z^N) < \mathrm{mult}_F \mathrm{div}(u^{4r})$$

for any F in the remaining cases of (i), and that

$$\mathrm{mult}_{F_r} \mathrm{div}(u^{4r}) < \mathrm{mult}_{F_r} \mathrm{div}(u^6 z^N).$$

The inequality for F_r is necessary for (iv), since a and b have equal orders along F_r, whereas $\mathrm{mult}_{F_r} \mathrm{div}(u^6 z^N)$ is odd.

(b) Two uses of the Kollár component condition. Without the Kollár component condition the bound fails: $\mathrm{mld}(X_r \ni 0) = \frac{2m}{4m-1}$, and a prime divisor over X_r computing it is the image of two prime divisors over Y_r exchanged by τ (Remark 5.2). Items (ii) and (iii) use twice that E is a Kollár component. First, E lifts to a single prime divisor F over Y_r (Definition-Proposition 2.6(2)). Thus F is G_r-invariant, the functions a and $b = -\tau^* a$ have equal orders along F, and

$$2\,\mathrm{mult}_F \mathrm{div}(a) = \mathrm{mult}_F \mathrm{div}(ab) = \mathrm{mult}_F \mathrm{div}(u^6 z^N)$$

is e times an odd integer. Hence e is even, and τ, the unique involution in the cyclic group G_r, acts trivially on F. Second, since F is a Kollár component, the total space of the Rees

construction of F over $\mathbb{A}^1$ and the fiber $\operatorname{Spec}\operatorname{gr}_F R$ over 0 form a plt pair [XZ25, Lemma 2.17], and hence $\operatorname{gr}_F R$ is normal (Definition-Lemma 2.5).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce Kollár components and the minimal log discrepancy of Kollár components, the graded algebra of a Kollár component, and the correspondence of Kollár components under finite quotients that are étale in codimension one. In Section 3 we write X_r as the quotient of Y_r by G_r, and prove that the quotient morphism is étale in codimension one and that $X_r \ni 0$ is klt. In Section 4 we prove that the log discrepancy of any Kollár component over $X_r \ni 0$ is at least $\frac{3m-1}{4m-1}$. In Section 5 we use a weighted blow-up of Y_r to construct a Kollár component over $X_r \ni 0$ whose log discrepancy is $\frac{3m-1}{4m-1}$, and prove Theorem 1.2.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and Kollár components.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

When $B = 0$, we write $a(E, X) := a(E, X, 0)$, and we write $X \ni x$ for the pair $(X \ni x, 0)$. If $(X \ni x, 0)$ is klt, then we say that $X \ni x$ is a *klt germ*.

Definition 2.2. Let $(X \ni x, B)$ be an lc pair, where x is a closed point. The *minimal log discrepancy* of $(X \ni x, B)$ is

$$\operatorname{mld}(X \ni x, B) := \min\{a(E, X, B) \mid E \text{ is a prime divisor over } X \text{ with } \operatorname{center}_X E = x\}.$$

When $B = 0$, we write $\operatorname{mld}(X \ni x) := \operatorname{mld}(X \ni x, 0)$.

Definition 2.3 ([Zhu24a, Definitions 1.4, 2.1, and 2.3])**.** Let $(X \ni x, B)$ be a klt pair, where x is a closed point. A prime divisor E over X with $\operatorname{center}_X E = x$ is called a *Kollár component* over $(X \ni x, B)$ if, possibly shrinking X to a neighborhood of x, there exists a projective birational morphism $f : Y \to X$ from a normal variety Y satisfying the following conditions: E is a prime divisor on Y, $-E$ is $\mathbb{Q}$-Cartier and f-ample, and $(Y, f_*^{-1}B_1 + E)$ is plt near E, where $B_1 := B - (\operatorname{mult}_E B)E$ if E is a divisor on X and $B_1 := B$ otherwise. The morphism f is uniquely determined by E [Zhu24a, Definition 2.1(1)], and we call f the *associated prime blow-up* of E. The *minimal log discrepancy of Kollár components* of $(X \ni x, B)$ is

$$\operatorname{mld}^{\mathrm{K}}(X \ni x, B) := \min\{a(E, X, B) \mid E \text{ is a Kollár component over } (X \ni x, B)\}.$$

When $B = 0$, we write $\mathrm{mld}^{\mathrm{K}}(X \ni x) := \mathrm{mld}^{\mathrm{K}}(X \ni x, 0)$.

Lemma 2.4. *Let $(X \ni x, B)$ be a klt pair, where x is a closed point and $\dim X \geq 2$, and let E be a Kollár component over $(X \ni x, B)$ with associated prime blow-up $f : Y \to X$. Then E is exceptional over X, f is an isomorphism over $X\backslash\{x\}$, $f^{-1}(x) = E$, and, possibly shrinking X to a neighborhood of x, $(Y, f_*^{-1}B + E)$ is plt. In particular, if B is a $\mathbb{Q}$-divisor, then E is a Kollár component over $(X \ni x, B)$ in the sense of* [LX20, Definition 1.1]*, and, if moreover X is affine, in the sense of* [XZ25, Definition 2.15]*. Conversely, if B is a $\mathbb{Q}$-divisor, then any Kollár component over $(X \ni x, B)$ in the sense of* [LX20, Definition 1.1] *is a Kollár component over $(X \ni x, B)$ in the sense of Definition 2.3.*

Proof. Since x is a closed point and $\dim X \geq 2$, the center $\mathrm{center}_X E = x$ has codimension at least two in X. Thus E is exceptional over X, and $B_1 = B$.

Since $-E$ is $\mathbb{Q}$-Cartier and f-ample, any curve $C \subset Y$ such that $f(C)$ is a point satisfies $(E \cdot C) < 0$, hence $C \subset E$. Since $f(E) = x$, f is quasi-finite over $X\backslash\{x\}$. Since X is normal and f is projective and birational, f is an isomorphism over $X\backslash\{x\}$ by Zariski's main theorem. By Zariski's main theorem again, $f^{-1}(x)$ is connected, and it contains E. Any irreducible component of $f^{-1}(x)$ of positive dimension is covered by curves C such that $f(C)$ is a point, hence is contained in E, and $f^{-1}(x)$ has no isolated point since it is connected and contains E. Therefore $f^{-1}(x) = E$.

Possibly shrinking X to a neighborhood of x, we may assume that (X, B) is klt. Since f is an isomorphism over $X\backslash\{x\}$, the pair

$$(Y\backslash E, f_*^{-1}B|_{Y\backslash E}) \cong (X\backslash\{x\}, B|_{X\backslash\{x\}})$$

is klt. Since $(Y, f_*^{-1}B + E)$ is also plt near E, $(Y, f_*^{-1}B + E)$ is plt.

Since E is the only exceptional prime divisor of f,

$$K_Y + f_*^{-1}B + E = f^*(K_X + B) + a(E, X, B)E,$$

and $a(E, X, B) > 0$ as $(X \ni x, B)$ is klt. Hence $-(K_Y + f_*^{-1}B + E) \sim_{\mathbb{R},X} -a(E, X, B)E$ is f-ample. Thus f is a proper birational morphism, $E = f^{-1}(x)$ is the unique exceptional divisor of f, $(Y, E + f_*^{-1}B)$ is plt, and $-(K_Y + E + f_*^{-1}B)$ is f-ample, which are the conditions of [XZ25, Definition 2.15] when B is a $\mathbb{Q}$-divisor and X is affine. Moreover, f is an isomorphism over $X\backslash\{x\}$, $f^{-1}(x)$ is the prime divisor E, $(Y, E + f_*^{-1}B)$ is plt, and $-E$ is $\mathbb{Q}$-Cartier and ample over X, which are the conditions of [LX20, Definition 1.1] when B is a $\mathbb{Q}$-divisor.

Conversely, assume that B is a $\mathbb{Q}$-divisor, and let $\mu : Y \to X$ be a proper birational morphism which provides a Kollár component S over $(X \ni x, B)$ in the sense of [LX20, Definition 1.1]. Since $-S$ is $\mathbb{Q}$-Cartier and ample over X, μ is projective. Since μ is an isomorphism over $X\backslash\{x\}$ and $\mu^{-1}(x) = S$, S is an exceptional prime divisor on the normal variety Y with $\mathrm{center}_X S = x$, and hence $B_1 = B$ in Definition 2.3. Since $(Y, \mu_*^{-1}B + S)$ is plt, S is a Kollár component over $(X \ni x, B)$ in the sense of Definition 2.3 with associated prime blow-up μ. □

2.2. **Graded algebras of Kollár components.** In Section 4, the plt condition in Definition 2.3 is used through the normality of the graded algebra $\mathrm{gr}_F R$ defined below.

Definition-Lemma 2.5 (cf. [XZ25, Lemmas 2.14 and 2.17])**.** Let $Y \ni y$ be a klt germ, $R := \mathcal{O}_{Y,y}$, and F a Kollár component over $Y \ni y$. For any integer $j \geq 0$, let

$$\mathfrak{a}_j(F) := \{0\} \cup \{\phi \in R\backslash\{0\} \mid \mathrm{mult}_F \mathrm{div}(\phi) \geq j\},$$

let

$$\mathrm{gr}_F R := \bigoplus_{j \geq 0} \mathfrak{a}_j(F)/\mathfrak{a}_{j+1}(F),$$

and for any $\phi \in R\backslash\{0\}$ with $\mathrm{mult}_F \mathrm{div}(\phi) = j$, let $\bar{\phi} \in (\mathrm{gr}_F R)_j$ be the image of ϕ. Then $\mathrm{gr}_F R$ is a finitely generated normal domain, $(\mathrm{gr}_F R)_0 = \mathbb{C}$, and $\overline{\phi\psi} = \bar{\phi}\bar{\psi}$ for any $\phi, \psi \in R\backslash\{0\}$.

Proof. The case $\dim Y = 1$ is trivial. We assume that $\dim Y \geq 2$.

For any $\phi, \psi \in R\backslash\{0\}$,

$$\operatorname{mult}_F \operatorname{div}(\phi\psi) = \operatorname{mult}_F \operatorname{div}(\phi) + \operatorname{mult}_F \operatorname{div}(\psi),$$

and

$$\operatorname{mult}_F \operatorname{div}(\phi + \psi) \geq \min\{\operatorname{mult}_F \operatorname{div}(\phi), \operatorname{mult}_F \operatorname{div}(\psi)\}$$

if $\phi+\psi \neq 0$. Thus any $\mathfrak{a}_j(F)$ is an ideal of R, $\mathfrak{a}_j(F)\mathfrak{a}_k(F) \subset \mathfrak{a}_{j+k}(F)$, $\operatorname{gr}_F R$ is a graded $\mathbb{C}$-algebra, and $\overline{\phi\psi} = \bar{\phi}\bar{\psi}$. In particular, the product of two non-zero homogeneous elements of $\operatorname{gr}_F R$ is non-zero. Comparing the non-zero homogeneous components of lowest degree, $\operatorname{gr}_F R$ is a domain. Since $\operatorname{center}_Y F = y$, a function $\phi \in R$ vanishes along F if and only if $\phi(y) = 0$. Hence $\mathfrak{a}_1(F)$ is the maximal ideal of R, and $(\operatorname{gr}_F R)_0 = \mathbb{C}$.

The ring in [XZ25] is the coordinate ring of an affine variety rather than a local ring. Let Y_0 be an affine neighborhood of y such that Y_0 is klt. Since $\operatorname{center}_Y F = y$, for any $j \geq 1$, the ideal of the regular functions on Y_0 vanishing to order at least j along F is primary to the maximal ideal of y. Hence replacing R by the coordinate ring of Y_0 does not change $\operatorname{gr}_F R$, and we may replace Y by Y_0, so that Y is affine and klt. Let $g : Y' \to Y$ be the associated prime blow-up of F. Since Y' is normal, g is projective and birational, and $-F$ is $\mathbb{Q}$-Cartier and g-ample, F is primitive in the sense of [XZ25, Section 2.4]. Thus $\operatorname{gr}_F R$ is finitely generated by [XZ25, Lemma 2.14]. Let $\mathcal{Y}$ be the total space of the Rees construction [XZ25, (2.1)] of F over $\mathbb{A}^1$, and $\mathcal{Y}_0$ the fiber over $0 \in \mathbb{A}^1$. Then $\mathcal{Y}_0 \cong \operatorname{Spec} \operatorname{gr}_F R$, which is a prime divisor on $\mathcal{Y}$ since $\operatorname{gr}_F R$ is a domain. By Lemma 2.4, F is a Kollár component over $Y \ni y$ in the sense of [XZ25, Definition 2.15]. By [XZ25, Lemma 2.17], $(\mathcal{Y}, \mathcal{Y}_0)$ is plt. By [KM98, Definitions 2.34 and 2.37], $(\mathcal{Y}, \mathcal{Y}_0)$ is dlt, where we take the closed subset in [KM98, Definition 2.37] to be the union of the singular loci of $\mathcal{Y}$ and $\mathcal{Y}_0$, which has codimension at least two in $\mathcal{Y}$. By [KM98, Proposition 5.51], $\mathcal{Y}_0$ is normal. Therefore $\operatorname{gr}_F R$ is normal. □

2.3. **Finite quotients.** The following statement is used in Sections 3, 4, and 5 for the finite cover of X_r constructed in Section 3.

Definition-Proposition 2.6 (cf. [LX20, Lemma 2.13])**.** Let $Y \ni y$ be a klt germ and G a finite group acting faithfully on Y with $\sigma(y) = y$ for any $\sigma \in G$. Let $q : Y \to X := Y/G$ be the quotient morphism and $x := q(y)$. Assume that q is étale in codimension one and $q^{-1}(x) = \{y\}$. We say that a prime divisor F over Y is *G-invariant* if

$$\operatorname{mult}_F \operatorname{div}(\sigma^*\phi) = \operatorname{mult}_F \operatorname{div}(\phi)$$

for any $\sigma \in G$ and any $\phi \in \mathcal{O}_{Y,y}\backslash\{0\}$. Then the following hold.

(1) $K_Y = q^*K_X$, K_X is $\mathbb{Q}$-Cartier near x, and $X \ni x$ is klt.
(2) Let E be a Kollár component over $X \ni x$ with associated prime blow-up $f : X' \to X$, let Y' be the normalization of the irreducible component of $X' \times_X Y$ dominating Y, and let $g : Y' \to Y$ and $q' : Y' \to X'$ be the induced morphisms. Then the action of G on Y lifts to Y', q' induces an isomorphism $Y'/G \cong X'$, $F := q'^{-1}(E)$ is a prime divisor, and F is a G-invariant Kollár component over $Y \ni y$ with associated prime blow-up g. Conversely, for any G-invariant Kollár component F over $Y \ni y$ with associated prime blow-up $g : Y' \to Y$, the action of G lifts to Y', and the image E of F in $X' := Y'/G$ is a Kollár component over $X \ni x$ with associated prime blow-up the induced morphism $X' \to X$.
(3) In the situation of (2), let $I_F \subset G$ be the subgroup of the elements σ whose lifts to Y' restrict to the identity on F, and let $e := |I_F|$. Then $q'^*E = eF$, $a(F, Y) = ea(E, X)$, and

$$\operatorname{mult}_F \operatorname{div}(q^*\phi) = e \operatorname{mult}_E \operatorname{div}(\phi)$$

for any non-zero regular function ϕ on a neighborhood of x.

(4) In the situation of (3), there exists an injective group homomorphism $\chi : I_F \to \mathbb{C}^*$ such that $\sigma^*\phi - \chi(\sigma)^j\phi \in \mathfrak{a}_{j+1}(F)$ for any $\sigma \in I_F$, any integer $j \geq 0$, and any $\phi \in \mathfrak{a}_j(F)$, where $\mathfrak{a}_j(F) \subset \mathcal{O}_{Y,y}$ is as in Definition-Lemma 2.5.

Proof. The case $\dim Y = 1$ is trivial. We assume that $\dim Y \geq 2$. Since q is finite and $q^{-1}(x) = \{y\}$, for any neighborhood V of y, the complement of $q(Y\backslash V)$ is a neighborhood of x whose preimage is contained in V. Possibly shrinking X to a neighborhood of x and Y to the preimage of the neighborhood, we may assume that Y is klt. In particular, K_Y is $\mathbb{R}$-Cartier with integer coefficients, hence $\mathbb{Q}$-Cartier.

(1) Since q is étale in codimension one, $K_Y = q^*K_X$ by the Hurwitz formula. By [KM98, Proposition 5.20(1)(4)] applied to q with zero boundaries, K_X is $\mathbb{Q}$-Cartier and X is klt. In particular, $X \ni x$ is klt.

(2) We first prove the first part of (2). By Lemma 2.4 applied to $X \ni x$, f is an isomorphism over $X\backslash\{x\}$ and $f^{-1}(x) = E$. Let W be the irreducible component of $X' \times_X Y$ dominating Y. The projection $X' \times_X Y \to Y$ is an isomorphism over $Y\backslash\{y\} = q^{-1}(X\backslash\{x\})$, and W is the closure of the preimage of $Y\backslash\{y\}$. Since Y is normal, g is an isomorphism over $Y\backslash\{y\}$. The group G acts on $X' \times_X Y$ through the second factor, and the action preserves W since $Y\backslash\{y\}$ is G-stable. By the universal property of normalization, the action of G lifts to Y', g is G-equivariant, and q' is G-invariant. Since $Y' \to W$ is finite and $X' \times_X Y \to X'$ is the base change of the finite morphism q, q' is finite. Hence the induced morphism $h : Y'/G \to X'$ is finite. Over $X'\backslash E \cong X\backslash\{x\}$, h is the isomorphism $(Y\backslash\{y\})/G \to X\backslash\{x\}$ induced by q, and hence h is birational. Since X' is normal, h is an isomorphism by Zariski's main theorem.

Since $f \circ q' = q \circ g$, $q^{-1}(x) = \{y\}$, and $f^{-1}(x) = E$, we have

$$q'^{-1}(E) = g^{-1}(y).$$

Since g is proper and birational and Y is normal, $q'^{-1}(E)$ is connected by Zariski's main theorem. Here g is proper since it is the composite of the finite morphism $Y' \to W$ and the proper morphism $W \to Y$, where W is closed in $X' \times_X Y$ and $X' \times_X Y \to Y$ is the base change of the projective morphism f. Since E is $\mathbb{Q}$-Cartier and q' is surjective, $q'^*E \geq 0$ is a non-zero $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor with support $q'^{-1}(E)$, and we write $q'^*E = \sum_i e_iT_i$, where the T_i are the irreducible components of $q'^{-1}(E)$ and $e_i > 0$. Since q' coincides with q over $X'\backslash E \cong X\backslash\{x\}$ and q is étale in codimension one, the Hurwitz formula implies

$$K_{Y'} + \sum_i T_i = q'^*(K_{X'} + E).$$

By (1), $K_{X'} + E = f^*K_X + a(E, X)E$ is $\mathbb{Q}$-Cartier. Since (X', E) is plt near E, [KM98, Proposition 5.20(3)] applied to q' over a neighborhood of E implies that $(Y', \sum_i T_i)$ is plt near $\sum_i T_i$. By [KM98, Definitions 2.34 and 2.37], $(Y', \sum_i T_i)$ is dlt near $\sum_i T_i$, where we take the closed subset in [KM98, Definition 2.37] to be the union of the singular locus of Y' and the locus where $\sum_i T_i$ is not a simple normal crossing divisor, which has codimension at least two in Y', and hence $\sum_i T_i$ is the disjoint union of the T_i by [KM98, Proposition 5.51]. Since $q'^{-1}(E)$ is connected, it has exactly one irreducible component. Thus $F = q'^{-1}(E)$ is a prime divisor.

Write $q'^*E = e'F$ for a positive rational number e'. Then $-F = -\frac{1}{e'}q'^*E$ is $\mathbb{Q}$-Cartier. Since $-E$ is f-ample, the pullback of $-E$ to $X' \times_X Y$ is ample over Y. Since $Y' \to X' \times_X Y$ is finite, $-F$ is g-ample, and g is projective. Moreover, $g(F) = y$ and (Y', F) is plt near F. Thus F is a Kollár component over $Y \ni y$ with associated prime blow-up g.

For any $\sigma \in G$, we denote the lift of σ to Y' also by σ. Since q' is G-invariant, σ preserves $q'^{-1}(E) = F$. Since g is G-equivariant, for any $\phi \in \mathcal{O}_{Y,y}\backslash\{0\}$, we have $g^*\sigma^*\phi = \sigma^*g^*\phi$, and hence

$$\mathrm{mult}_F \operatorname{div}(\sigma^*\phi) = \mathrm{mult}_{\sigma(F)} \operatorname{div}(g^*\phi) = \mathrm{mult}_F \operatorname{div}(\phi).$$

Thus F is G-invariant.

We prove the converse part of (2). Let F be a G-invariant Kollár component over $Y \ni y$ with associated prime blow-up $g : Y' \to Y$. By Lemma 2.4 applied to $Y \ni y$, F is a Kollár

component over $Y \ni y$ in the sense of [LX20, Definition 1.1]. Since q is a Galois covering with Galois group G and $q^*K_X = K_Y$ by (1), [LX20, Lemma 2.13] implies that F is the pullback of a Kollár component E over $X \ni x$ in the sense of [LX20, Definition 1.1]. By Lemma 2.4 applied to $X \ni x$, E is a Kollár component over $X \ni x$ in the sense of Definition 2.3. Let $X'_E \to X$ be the associated prime blow-up of E, and let $g_E : Y'_E \to Y$ and $q'_E : Y'_E \to X'_E$ be the morphisms constructed from E in the first part of (2). Since F is the pullback of E in the sense of [LX20, Lemma 2.13], the prime divisor $q'^{-1}_E(E)$ on Y'_E is F (since q'_E is finite, $q'^{-1}_E(E)$ is the only prime divisor over Y lying over E). By the first part of (2), F is a Kollár component over $Y \ni y$ with associated prime blow-up g_E. Since the associated prime blow-up is uniquely determined by F [Zhu24a, Definition 2.1(1)], g_E is g. Thus, by the first part of (2), the action of G lifts to Y', $Y'/G \cong X'_E$ is the associated prime blow-up of E, and E is the image of F in Y'/G.
(3) By (2), q' is the quotient morphism of the action of G on Y', and the action is faithful since g is G-equivariant and birational. Hence $\deg q' = |G|$, and by the projection formula,

$$q'_* q'^* E = |G| E.$$

By (2), $q'^{-1}(E) = F$ is a prime divisor, and hence $q'^* E = e' F$ for a positive rational number e' (the ramification index of q' along F). Then

$$q'_* q'^* E = e' \deg(q'|_F) E.$$

The fibers of q' are G-orbits, and G/I_F acts faithfully on F. Hence the orbit of a general point of F has $|G|/|I_F|$ points, and since we work in characteristic zero, $\deg(q'|_F) = |G|/|I_F|$. Therefore $e' = |I_F| = e$, that is, $q'^* E = eF$.

For any non-zero regular function ϕ on a neighborhood of x, $g^* q^* \phi = q'^* f^* \phi$. Since $q'^* E = eF$, the coefficient of F in $q'^* \operatorname{div}(f^*\phi)$ is $e \operatorname{mult}_E \operatorname{div}(f^*\phi)$. Thus

$$\operatorname{mult}_F \operatorname{div}(q^*\phi) = e \operatorname{mult}_E \operatorname{div}(\phi).$$

Since q' coincides with q over $X' \backslash E \cong X \backslash \{x\}$, q is étale in codimension one, and $q'^* E = eF$, the Hurwitz formula implies

$$K_{Y'} = q'^* K_{X'} + (e-1)F.$$

Since $K_{X'} = f^* K_X + (a(E,X) - 1)E$ and $K_Y = q^* K_X$,

$$K_{Y'} = q'^* f^* K_X + e(a(E,X) - 1)F + (e-1)F = g^* K_Y + (ea(E,X) - 1)F.$$

Comparing the coefficients of F with $K_{Y'} = g^* K_Y + (a(F,Y) - 1)F$, we obtain $a(F,Y) = ea(E,X)$.
(4) Let $p \in F$ be a general point, so that Y' and F are smooth at p, and let w be a local equation of F near p. We denote the lift of any $\sigma \in G$ to Y' also by σ. For $\sigma \in I_F$, σ preserves F and fixes p, hence $\sigma^* w = c_\sigma w$ for a function c_σ invertible near p. Let $\chi(\sigma) := c_\sigma|_F$. Replacing w by vw for an invertible function v multiplies c_σ by $\sigma^* v / v$, which is 1 on F. Hence $\chi(\sigma)$ does not depend on w. For $\sigma, \tau \in I_F$, $(\sigma\tau)^* w = \tau^*(c_\sigma) c_\tau w$ and $\tau^*(c_\sigma)|_F = c_\sigma|_F$, so $\chi(\sigma\tau) = \chi(\sigma)\chi(\tau)$. Since $\chi(\sigma)^{|I_F|} = 1$, $\chi(\sigma)$ is a regular function on an open subset of F taking finitely many values. Since F is irreducible, $\chi(\sigma) \in \mathbb{C}^*$ is a constant independent of p, and $\chi : I_F \to \mathbb{C}^*$ is a group homomorphism.

Let $\sigma \in I_F$, $j \geq 0$ an integer, and $\phi \in \mathfrak{a}_j(F)$. Near p, write $g^*\phi = w^j \phi_1$ with ϕ_1 regular. Then

$$g^*(\sigma^*\phi - \chi(\sigma)^j \phi) = w^j \big((c_\sigma^j - \chi(\sigma)^j) \sigma^* \phi_1 + \chi(\sigma)^j (\sigma^* \phi_1 - \phi_1) \big),$$

and both $c_\sigma^j - \chi(\sigma)^j$ and $\sigma^*\phi_1 - \phi_1$ vanish along F. Since $\sigma(y) = y$, $\sigma^*\phi - \chi(\sigma)^j \phi \in \mathcal{O}_{Y,y}$, and hence $\sigma^*\phi - \chi(\sigma)^j \phi \in \mathfrak{a}_{j+1}(F)$.

Finally, let $\sigma \in I_F$ with $\chi(\sigma) = 1$, and let n be the order of σ. Suppose that there exists $\phi \in \mathcal{O}_{Y,y}$ such that $\psi := \sigma^*\phi - \phi \neq 0$, and let $k := \operatorname{mult}_F \operatorname{div}(\psi)$. Since $\chi(\sigma^i) = 1$, $(\sigma^i)^*\psi - \psi \in \mathfrak{a}_{k+1}(F)$ for any $0 \leq i < n$. Since

$$0 = (\sigma^n)^*\phi - \phi = \sum_{i=0}^{n-1} (\sigma^i)^* \psi,$$

we have $n\psi \in \mathfrak{a}_{k+1}(F)$, which contradicts $\operatorname{mult}_F \operatorname{div}(n\psi) = k$. Thus σ^* is the identity on $\mathcal{O}_{Y,y}$, and hence σ is the identity on Y. Since G acts faithfully on Y, $\sigma = 1$. Therefore χ is injective. □

3. Construction of the singularities

In this section, we prove Lemma 3.3, by which the quotient morphism $q_r : Y_r \to X_r$ of Construction 3.1 satisfies the hypotheses of Definition-Proposition 2.6.

Construction 3.1. Let $m \geq 2$ be an integer, $r := 4m - 1$, and $N := 2r + 3$. Let $\mu_r \subset \mathbb{C}^*$ be the group of r-th roots of unity.

(1) Let X'_r be the hypersurface $(t^2 = y(x^2 + y^{2r} + y^3 z^N))$ in $\mathbb{A}^4_{x,y,z,t}$, with the action of μ_r given by $\zeta \cdot (x, y, z, t) = (x, \zeta y, \zeta^{-1} z, \zeta^{2m} t)$. Let $X_r := X'_r/\mu_r$, and let $0 \in X_r$ be the image of the origin.

(2) Let Y_r be the hypersurface $(ab = u^{4r} + u^6 z^N)$ in $\mathbb{A}^4_{a,b,u,z}$, and let $0 \in Y_r$ be the origin. Let τ be the involution of Y_r given by $\tau(a, b, u, z) = (-b, -a, -u, z)$, and let μ_r act on Y_r by $\zeta \cdot (a, b, u, z) = (a, b, \zeta^{2m} u, \zeta^{-1} z)$. Let G_r be the group of automorphisms of Y_r generated by τ and μ_r, and let $q_r : Y_r \to Y_r/G_r$ be the quotient morphism.

Lemma 3.2(3) below bounds from below the log discrepancy of any prime divisor over the hypersurface $(ab = \phi) \subset \mathbb{A}^4_{a,b,u,z}$ in terms of the orders of u and z, and its proof only uses the generator of the canonical sheaf in Lemma 3.2(2). It is the starting point of the lower bound in Section 4. We state Lemma 3.2 for a class of polynomials ϕ, since besides the case $\phi = u^{4r} + u^6 z^N$, the case $\phi = u^{4r}$ is used in the proof of Construction-Proposition 5.1(1) in Section 5, where $(ab = u^{4r})$ needs to have canonical singularities.

Lemma 3.2. *Let $\phi \in \mathbb{C}[u, z]$ be a non-zero polynomial such that ϕ, $\partial\phi/\partial u$, and $\partial\phi/\partial z$ have no common zero on $\mathbb{A}^2_{u,z} \setminus (u = 0)$, and let $Y := (ab = \phi(u, z)) \subset \mathbb{A}^4_{a,b,u,z}$. Then the following hold.*

(1) Y is a normal variety with canonical singularities.

(2) The canonical sheaf ω_Y is generated by $a^{-1} da \wedge du \wedge dz$. In particular, $K_Y \sim 0$.

(3) For any prime divisor F over Y,

$$a(F, Y) \geq \operatorname{mult}_F \operatorname{div}(u) + \operatorname{mult}_F \operatorname{div}(z).$$

Proof. We first prove that Y in (1) is a normal variety. The polynomial $ab - \phi$ has degree one in a, and the coefficients b and $-\phi$ have no common factor in $\mathbb{C}[b, u, z]$, since $\phi \neq 0$ does not involve b. Thus $ab - \phi$ is irreducible, Y is a variety, and $\dim Y = 3$. A point of Y is singular if and only if

$$a = b = \frac{\partial \phi}{\partial u} = \frac{\partial \phi}{\partial z} = 0$$

at the point. Since $ab = \phi$ on Y, ϕ also vanishes at a singular point, so $u = 0$ at a singular point by assumption. Hence

$$\operatorname{Sing}(Y) \subset (a = b = u = 0),$$

and $\operatorname{codim}_Y \operatorname{Sing}(Y) \geq 2$. Since Y is a hypersurface in $\mathbb{A}^4$ which is regular in codimension one, Y is normal by [Har77, Proposition II.8.23(b)].

Next we prove (2). Let $\Omega := a^{-1} da \wedge du \wedge dz$, a rational 3-form on Y. On $Y \cap (a \neq 0)$, we have $b = \phi/a$, so the projection to the coordinates (a, u, z) is an isomorphism from $Y \cap (a \neq 0)$ onto $(a \neq 0) \subset \mathbb{A}^3_{a,u,z}$, and Ω is regular and nowhere zero on $Y \cap (a \neq 0)$. Differentiating $ab = \phi$, we obtain $b\,da + a\,db = d\phi$. Since $d\phi$ is a combination of du and dz, $d\phi \wedge du \wedge dz = 0$, hence

$$b\,da \wedge du \wedge dz = -a\,db \wedge du \wedge dz, \qquad \text{that is,} \qquad \Omega = -b^{-1} db \wedge du \wedge dz.$$

Exchanging the roles of a and b, Ω is regular and nowhere zero on $Y \cap (b \neq 0)$. Since $\phi \neq 0$, the complement

$$Y \cap (a = b = 0) = (a = b = \phi = 0)$$

is of dimension at most 1, so it is of codimension at least 2 in Y. Since Y is normal, $\omega_Y = \mathcal{O}_Y \cdot \Omega$, and $K_Y = \operatorname{div}(\Omega) = 0$.

Next we prove (3). Let $f : Y' \to Y$ be a birational morphism from a normal variety Y' such that F is a prime divisor on Y', and let

$$c_1 := \operatorname{mult}_F \operatorname{div}(a), \quad c_2 := \operatorname{mult}_F \operatorname{div}(u), \quad c_3 := \operatorname{mult}_F \operatorname{div}(z).$$

Since $K_Y = \operatorname{div}(\Omega) = 0$ by (2), we have

$$a(F, Y) = 1 + \operatorname{mult}_F \operatorname{div}(f^*\Omega)$$

(the log discrepancy depends only on the valuation of F, so it can be computed on any normal model on which F is a divisor [KM98, Remark 2.23]). Let y' be a general point of F. Then Y' is smooth near y'. Let s be a local equation of F near y'. There exist regular functions a_1, u_1, and z_1 near y', invertible at y', such that $f^*a = s^{c_1}a_1$, $f^*u = s^{c_2}u_1$, and $f^*z = s^{c_3}z_1$. Then, near y',

$$f^*\Omega = s^{c_2+c_3}u_1z_1\Big(c_1\frac{ds}{s} + \frac{da_1}{a_1}\Big) \wedge \Big(c_2\frac{ds}{s} + \frac{du_1}{u_1}\Big) \wedge \Big(c_3\frac{ds}{s} + \frac{dz_1}{z_1}\Big).$$

Since $ds \wedge ds = 0$, after expanding the product, any non-zero term contains at most one factor $s^{-1}ds$. Thus $f^*\Omega = s^{c_2+c_3-1}\eta$ for some regular 3-form η near y', so

$$\operatorname{mult}_F \operatorname{div}(f^*\Omega) \geq c_2 + c_3 - 1.$$

Hence $a(F, Y) \geq c_2 + c_3$, which is (3).

Finally we prove that Y has canonical singularities. By (2), K_Y is Cartier. Let F be an exceptional prime divisor over Y with center V on Y. If $V \subset \operatorname{Sing}(Y)$, then u vanishes on V as $\operatorname{Sing}(Y) \subset (u = 0)$, and z is regular near the generic point of V. Thus

$$\operatorname{mult}_F \operatorname{div}(u) \geq 1 \quad \text{and} \quad \operatorname{mult}_F \operatorname{div}(z) \geq 0,$$

and $a(F, Y) \geq 1$ by (3). If $V \not\subset \operatorname{Sing}(Y)$, then F is an exceptional prime divisor over the smooth variety $Y \setminus \operatorname{Sing}(Y)$, so $a(F, Y) \geq 2$ by [KM98, Corollary 2.31(2)]. Therefore, $a(F, Y) \geq 1$ for any exceptional prime divisor F over Y, and Y has canonical singularities. □

Lemma 3.3. *Notation as in Construction 3.1. Then the following hold.*

1. *The assumption of Lemma 3.2 holds for $\phi = u^{4r} + u^6z^N$. In particular, $Y_r \ni 0$ is a klt germ.*
2. *τ commutes with the action of μ_r, G_r is a cyclic group of order $2r$, and τ is the unique element of order 2 in G_r.*
3. *The morphism $Y_r \to \mathbb{A}^4_{x,y,z,t}$ given by $(x, y, z, t) = ((b - a)/2, u^2, z, u(a + b)/2)$ induces an isomorphism $Y_r/\langle\tau\rangle \cong X'_r$ compatible with the actions of μ_r. In particular, X'_r is normal, $Y_r/G_r \cong X_r$, and $q_r(0) = 0$.*
4. *q_r is étale in codimension one and $q_r^{-1}(0) = \{0\}$.*
5. *$K_{Y_r} = q_r^*K_{X_r}$, K_{X_r} is $\mathbb{Q}$-Cartier, and $X_r \ni 0$ is klt.*

Proof. (1) Let $\phi := u^{4r} + u^6z^N$. Then $\partial\phi/\partial z = Nu^6z^{N-1}$. If $u \neq 0$ and $\partial\phi/\partial z = 0$, then $z = 0$, and $\phi = u^{4r} \neq 0$. Thus ϕ, $\partial\phi/\partial u$, and $\partial\phi/\partial z$ have no common zero on $\mathbb{A}^2_{u,z}\setminus(u = 0)$, and the assumption of Lemma 3.2 holds. By Lemma 3.2(1), Y_r is a normal variety with canonical singularities. In particular, $Y_r \ni 0$ is a klt germ.

(2) We first check that μ_r and τ preserve Y_r and that μ_r preserves X'_r. For any $\zeta \in \mu_r$, we have $(\zeta^{2m})^{4r} = 1$ and

$$(\zeta^{2m})^6\zeta^{-N} = \zeta^{12m-N} = \zeta^r = 1,$$

so ζ preserves ab, u^{4r}, and u^6z^N. The involution τ also preserves ab, u^{4r}, and u^6z^N, since $(-b)(-a) = ab$ and $4r$ and 6 are even. Thus μ_r and τ preserve Y_r. Since $\zeta^{4m} = \zeta$ and $\zeta^{2r} = \zeta^{3-N} = 1$, ζ multiplies both t^2 and $y(x^2 + y^{2r} + y^3z^N)$ by ζ, so μ_r preserves X'_r.

For any $\zeta \in \mu_r$ and any point $(a, b, u, z) \in Y_r$,

$$\tau(\zeta \cdot (a, b, u, z)) = (-b, -a, -\zeta^{2m}u, \zeta^{-1}z) = \zeta \cdot \tau(a, b, u, z),$$

so τ commutes with the action of μ_r. Thus $\mu_r \times \langle\tau\rangle \to G_r$ is a surjective group homomorphism. It is injective: an element (ζ, τ^i) acting trivially on Y_r acts trivially on z, so $\zeta = 1$, and $\tau \neq \mathrm{id}_{Y_r}$. Hence

$$G_r \cong \mu_r \times \mathbb{Z}/2\mathbb{Z},$$

which is cyclic of order $2r$ since r is odd. In particular, τ is the unique element of order 2 in G_r.

(3) Let $x := (b-a)/2$ and $v := (a+b)/2$. Then $R := \mathbb{C}[a,b,u,z] = \mathbb{C}[x,v,u,z]$, $ab = v^2 - x^2$, and $\mathbb{C}[Y_r] = R/(\psi)$, where

$$\psi := v^2 - x^2 - u^{4r} - u^6 z^N.$$

The involution τ fixes x and z, changes the signs of u and v, and fixes ψ. Let $\rho(g) := (g + \tau^* g)/2$ be the Reynolds operator on R. If the class of $g \in R$ in $R/(\psi)$ is τ-invariant, then $\rho(g) \in R^\tau$ and $\rho(g) - g \in (\psi)$. If $\psi g \in R^\tau$, then $\psi g = \rho(\psi g) = \psi \rho(g)$. Thus taking τ-invariants commutes with taking the quotient by ψ, that is,

$$\mathbb{C}[Y_r]^\tau = R^\tau/\psi R^\tau.$$

The ring R^τ is generated by x, z, $y := u^2$, $t := uv$, and v^2, and the ideal of relations among them is generated by $t^2 - yv^2$, since the τ-invariant part of $\mathbb{C}[u,v]$ is the even-degree part $\mathbb{C}[u^2, uv, v^2]$. Since $\psi = v^2 - x^2 - y^{2r} - y^3 z^N$, eliminating v^2, we obtain

$$\mathbb{C}[Y_r]^\tau \cong \mathbb{C}[x,y,z,t]/\big(t^2 - y(x^2 + y^{2r} + y^3 z^N)\big).$$

Thus the morphism in (3) induces an isomorphism $Y_r/\langle\tau\rangle \cong X'_r$. For any $\zeta \in \mu_r$, the morphism sends $\zeta \cdot (a,b,u,z)$ to

$$(x, \zeta^{4m} u^2, \zeta^{-1} z, \zeta^{2m} uv) = (x, \zeta y, \zeta^{-1} z, \zeta^{2m} t) = \zeta \cdot (x,y,z,t),$$

since $4m = r+1$. Thus the isomorphism is compatible with the actions of μ_r. By (1), $\mathbb{C}[Y_r]$ is a normal domain. Since

$$\mathbb{C}[Y_r]^\tau = \mathbb{C}[Y_r] \cap \mathrm{Frac}(\mathbb{C}[Y_r])^\tau,$$

$\mathbb{C}[Y_r]^\tau$ is normal, so X'_r is normal. Since $G_r = \mu_r \times \langle\tau\rangle$ by (2),

$$Y_r/G_r \cong (Y_r/\langle\tau\rangle)/\mu_r \cong X'_r/\mu_r = X_r.$$

The origin of Y_r maps to the origin of X'_r, so $q_r(0) = 0$.

(4) By (2), any element of $G_r \backslash \{1\}$ is either ζ or $\tau\zeta$ for some $\zeta \in \mu_r$. For any $\zeta \in \mu_r \backslash \{1\}$, we have $\zeta^{2m} \neq 1$ since $2m$ and $r = 4m-1$ are coprime, so a fixed point of ζ satisfies $u = z = 0$. For any $\zeta \in \mu_r$, we have $-\zeta^{2m} \neq 1$ since $(-\zeta^{2m})^r = -1$ as r is odd, so a fixed point of $\tau\zeta$ satisfies $u = 0$ and $a = -b$. Then the equation of Y_r implies $-a^2 = ab = 0$, so $a = b = 0$. Thus the fixed locus of any element of $G_r \backslash \{1\}$ is contained in

$$Y_r \cap (u = z = 0) \quad \text{or} \quad (a = b = u = 0),$$

both of which are of dimension 1, hence of codimension 2 in Y_r. Hence q_r is étale in codimension one (since the quotient morphism by a finite group is étale over the locus where the group acts freely). Since the fibers of q_r are G_r-orbits and G_r fixes the origin, $q_r^{-1}(0) = \{0\}$ by (3).

(5) By (1), $Y_r \ni 0$ is a klt germ, and G_r acts faithfully on Y_r and fixes 0. By (3) and (4), $q_r(0) = 0$, q_r is étale in codimension one, and $q_r^{-1}(0) = \{0\}$. Thus, by Definition-Proposition 2.6(1), $K_{Y_r} = q_r^* K_{X_r}$ and $X_r \ni 0$ is klt. Moreover, $K_{Y_r} \sim 0$ by Lemma 3.2(2), whose assumption holds by (1). Since q_r is finite, K_{Y_r} is Cartier, and $K_{Y_r} = q_r^* K_{X_r}$, K_{X_r} is $\mathbb{Q}$-Cartier by [KM98, Proposition 5.20(1)]. □

Remark 3.4. Notation as in Construction 3.1. Let μ_r act on $\mathbb{A}^3_{x,y,z}$ by $\zeta \cdot (x,y,z) = (x, \zeta y, \zeta^{-1} z)$, let $W_r := \mathbb{A}^3_{x,y,z}/\mu_r$, and let D_r and H_r be the images of $(x^2 + y^{2r} + y^3 z^N = 0)$ and $(y = 0)$ in W_r respectively. Then the morphism $X'_r \to \mathbb{A}^3_{x,y,z}$ forgetting t induces a finite morphism $g : X_r \to W_r$ of degree 2, and

$$K_{X_r} = g^*\Big(K_{W_r} + \frac{1}{2} D_r + \frac{1}{2} H_r\Big).$$

Indeed, let $f : X'_r \to \mathbb{A}^3_{x,y,z}$ be the morphism forgetting t, and let $\psi := x^2 + y^{2r} + y^3 z^N$. Then f is the quotient by the involution of X'_r changing the sign of t, and $\deg f = 2$. The polynomial ψ is monic of degree 2 in x, and $y^{2r} + y^3 z^N = y^3(y^{2r-3} + z^N)$ is not a square in $\mathbb{C}[y, z]$, so ψ is irreducible and $y \nmid \psi$. Thus f is étale over $\mathbb{A}^3 \backslash (y\psi = 0)$, and $f^*(y = 0) = 2(t = y = 0)$ and $f^*(\psi = 0) = 2(t = \psi = 0)$ with $(t = y = 0)$ and $(t = \psi = 0)$ prime divisors on X'_r, so the ramification formula implies

$$K_{X'_r} = f^*\Big(K_{\mathbb{A}^3} + \frac{1}{2}(\psi = 0) + \frac{1}{2}(y = 0)\Big).$$

Let $h : X'_r \to X_r$ and $h' : \mathbb{A}^3_{x,y,z} \to W_r$ be the quotient morphisms, so that $g \circ h = h' \circ f$. In particular, g is finite as $h' \circ f$ is finite, and $\deg g = 2$ as $\deg f = 2$ and $\deg h = \deg h' = r$. For any $\zeta \in \mu_r \backslash \{1\}$, the fixed points of ζ on $\mathbb{A}^3_{x,y,z}$ and on X'_r are contained in $(y = z = 0)$, so h and h' are étale in codimension one (since the quotient morphism by a finite group is étale over the locus where the group acts freely). Hence $K_{X'_r} = h^* K_{X_r}$, $K_{\mathbb{A}^3} = h'^* K_{W_r}$, $h'^* D_r = (\psi = 0)$, and $h'^* H_r = (y = 0)$, and therefore

$$h^* K_{X_r} = h^* g^*\Big(K_{W_r} + \frac{1}{2} D_r + \frac{1}{2} H_r\Big).$$

Since $h_* h^* = r \cdot \mathrm{id}$ on $\mathbb{Q}$-divisors, the equality for K_{X_r} follows. In particular, g is a double cover branched along $D_r + H_r$. We will not use Remark 3.4 in the rest of this paper.

4. A lower bound for log discrepancies of Kollár components

In this section, we prove Proposition 4.1, which is the lower bound for $\mathrm{mld}^{\mathrm{K}}(X_r \ni 0)$ in Theorem 1.2.

Proposition 4.1. *Notation as in Construction 3.1. Then $a(E, X_r) \geq \frac{3m-1}{4m-1}$ for any Kollár component E over $X_r \ni 0$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we replace E with a G_r-invariant Kollár component F over $Y_r \ni 0$ and obtain divisibility relations for the order e of I_F.

Let E be a Kollár component over $X_r \ni 0$. The group G_r acts faithfully on Y_r and fixes 0. By Lemma 3.3(1)(3)(4), $Y_r \ni 0$ is a klt germ, $Y_r/G_r \cong X_r$, $q_r(0) = 0$, q_r is étale in codimension one, and $q_r^{-1}(0) = \{0\}$. Thus the assumptions of Definition-Proposition 2.6 hold for $Y := Y_r$, $y := 0$, $G := G_r$, and $q := q_r$, with $X = X_r$ and $x = 0$. Let F be the G_r-invariant Kollár component over $Y_r \ni 0$ corresponding to E as in Definition-Proposition 2.6(2), let $I_F \subset G_r$ be as in Definition-Proposition 2.6(3), and let $e := |I_F|$. By Definition-Proposition 2.6(3),

$$a(E, X_r) = \frac{a(F, Y_r)}{e}, \tag{4.1}$$

and

$$\mathrm{mult}_F \operatorname{div}(q_r^* \phi) = e \,\mathrm{mult}_E \operatorname{div}(\phi)$$

for any non-zero regular function ϕ on a neighborhood of $0 \in X_r$.

Let

$$p := \mathrm{mult}_F \operatorname{div}(a), \quad c := \mathrm{mult}_F \operatorname{div}(u), \quad d := \mathrm{mult}_F \operatorname{div}(z).$$

Since $\mathrm{center}_{Y_r} F = 0$ and a, u, z vanish at 0, $p, c, d > 0$. Since $\tau^* a = -b$ and F is G_r-invariant,

$$\mathrm{mult}_F \operatorname{div}(b) = \mathrm{mult}_F \operatorname{div}(\tau^* a) = p.$$

For any $\zeta \in \mu_r$, $\zeta^{-r} = 1$ and $(\zeta^{2m})^2 \zeta^{-1} = \zeta^{4m-1} = 1$. Moreover, $\tau^* u = -u$ and $\tau^* z = z$. Thus z^r and $u^2 z$ are G_r-invariant regular functions on Y_r, hence they are pullbacks of non-zero regular functions on $X_r \cong Y_r/G_r$. Therefore e divides $\mathrm{mult}_F \operatorname{div}(z^r) = rd$ and $\mathrm{mult}_F \operatorname{div}(u^2 z) = 2c + d$. Let

$$k := \frac{rd}{e}, \quad s := \frac{2c + d}{e}.$$

Then k and s are positive integers.

Step 2. In this step, we reduce to the case when $s = 1$ and $1 \le k \le \frac{r-3}{2}$, which consists of $2m-2$ values of k.

By Lemma 3.3(1) and Lemma 3.2(3), $a(F, Y_r) \ge c + d$. By (4.1),

$$a(E, X_r) \ge \frac{c+d}{e} = \frac{2c+d}{2e} + \frac{d}{2e} = \frac{s}{2} + \frac{k}{2r}. \tag{4.2}$$

If $s \ge 2$, then (4.2) implies that $a(E, X_r) > 1 > \frac{3m-1}{4m-1}$. If $s = 1$ and $k \ge \frac{r-1}{2}$, then (4.2) implies that

$$a(E, X_r) \ge \frac{1}{2} + \frac{r-1}{4r} = \frac{3r-1}{4r} = \frac{3m-1}{4m-1},$$

where the last equality follows from $r = 4m-1$. Thus we may assume that $s = 1$ and $k < \frac{r-1}{2}$. Since r is odd, $\frac{r-1}{2}$ is an integer, hence $1 \le k \le \frac{r-3}{2}$.

Suppose that $s = 1$ and $1 \le k \le \frac{r-3}{2}$. Since $s = 1$, $2c = e - d$. Since $d = \frac{ek}{r}$,

$$c = \frac{e(r-k)}{2r}, \quad d = \frac{ek}{r}. \tag{4.3}$$

Step 3. In this step, we show that e is even and $\tau \in I_F$.

Since $N = 2r + 3$, (4.3) implies that

$$\operatorname{mult}_F \operatorname{div}(u^{4r}) = 4rc = e(2r - 2k),$$

$$\operatorname{mult}_F \operatorname{div}(u^6 z^N) = 6c + Nd = \frac{3e(r-k) + (2r+3)ek}{r} = e(3 + 2k).$$

Since $k \le \frac{r-3}{2}$,

$$e(2r - 2k) - e(3 + 2k) = e(2r - 3 - 4k) \ge 3e > 0.$$

Thus $\operatorname{mult}_F \operatorname{div}(u^{4r} + u^6 z^N) = e(3 + 2k)$. Since $ab = u^{4r} + u^6 z^N$ on Y_r,

$$2p = e(3 + 2k) = 6c + Nd. \tag{4.4}$$

Since $3 + 2k$ is odd, e is even. By Lemma 3.3(2), G_r is a cyclic group of order $2r$ and τ is the unique element of order 2 in G_r. Since I_F is a subgroup of G_r of even order e, I_F contains an element of order 2, hence $\tau \in I_F$.

Let $R := \mathcal{O}_{Y_r,0}$, let $\mathfrak{a}_j(F) \subset R$ be as in Definition-Lemma 2.5, and let $\chi : I_F \to \mathbb{C}^*$ be as in Definition-Proposition 2.6(4). Since χ is injective and τ is of order 2, $\chi(\tau) = -1$. By Definition-Proposition 2.6(4),

$$\tau^* \phi - (-1)^j \phi \in \mathfrak{a}_{j+1}(F) \tag{4.5}$$

for any integer $j \ge 0$ and any $\phi \in \mathfrak{a}_j(F)$.

Step 4. In this step, we derive a contradiction from the normality of $\operatorname{gr}_F R$.

By Lemma 3.3(1) and Definition-Lemma 2.5, $\operatorname{gr}_F R$ is a finitely generated normal domain and $(\operatorname{gr}_F R)_0 = \mathbb{C}$. Since $\tau^* a = -b$, (4.5) for $\phi = a$ and $j = p$ implies that $b + (-1)^p a \in \mathfrak{a}_{p+1}(F)$. Since $\operatorname{mult}_F \operatorname{div}(a) = \operatorname{mult}_F \operatorname{div}(b) = p$,

$$\bar{b} = (-1)^{p+1} \bar{a}.$$

By **Step 3**, $\operatorname{mult}_F \operatorname{div}(u^{4r}) > e(3 + 2k) = 2p$, hence $ab - u^6 z^N = u^{4r} \in \mathfrak{a}_{2p+1}(F)$. Since

$$\operatorname{mult}_F \operatorname{div}(ab) = \operatorname{mult}_F \operatorname{div}(u^6 z^N) = 2p,$$

we have $\overline{ab} = \overline{u^6 z^N}$. By Definition-Lemma 2.5,

$$(-1)^{p+1} \bar{a}^2 = \bar{a}\bar{b} = \bar{u}^6 \bar{z}^{2r+3}.$$

Let $\kappa \in \mathbb{C}$ be a complex number with $\kappa^2 = (-1)^{p+1}$, and let $W := \bar{u}^3 \bar{z}^{r+1}$. Since $\operatorname{gr}_F R$ is a domain, $W \ne 0$, and

$$(\kappa \bar{a})^2 = W^2 \bar{z}.$$

Thus the element $\kappa\bar{a}/W$ of the fraction field of $\mathrm{gr}_F R$ is a root of the monic polynomial $T^2-\bar{z}$ over $\mathrm{gr}_F R$. Since $\mathrm{gr}_F R$ is normal, there exists $Q\in \mathrm{gr}_F R$ such that $\kappa\bar{a}=QW$. Since $(QW)^2=(\kappa\bar{a})^2=W^2\bar{z}$ and $\mathrm{gr}_F R$ is a domain, $Q^2=\bar{z}$. Since $\mathrm{gr}_F R$ is a domain and $\kappa\bar{a}$ and W are non-zero homogeneous elements, Q is a non-zero homogeneous element (compare the homogeneous components of QW). By (4.4) and $N=2r+3$,

$$\deg Q = p-3c-(r+1)d = \frac{6c+(2r+3)d}{2}-3c-(r+1)d = \frac{d}{2}>0. \tag{4.6}$$

The maximal ideal $\mathfrak{m}_R$ of R is generated by a, b, u, and z. Since $\mathrm{center}_{Y_r} F=0$, $\mathrm{mult}_F \mathrm{div}(g)\geq 0$ for any $g\in R\backslash\{0\}$. Let j be a positive integer and θ a non-zero element of $(\mathrm{gr}_F R)_j$. Then $\theta=\bar{\phi}$ for some $\phi\in R\backslash\{0\}$ with $\mathrm{mult}_F \mathrm{div}(\phi)=j$. Since $j>0$, ϕ is not a unit of R, hence $\phi\in\mathfrak{m}_R$. Thus $\phi=g_1a+g_2b+g_3u+g_4z$ for some $g_1,g_2,g_3,g_4\in R$, and

$$j=\mathrm{mult}_F \mathrm{div}(\phi)\geq \min\{p,c,d\}.$$

By (4.6), $\frac{d}{2}\geq\min\{p,c,d\}$. On the other hand, $\frac{d}{2}<d$. Since $2k\leq r-3<r$, (4.3) implies that

$$\frac{d}{2}=\frac{ek}{2r}<\frac{e(r-k)}{2r}=c.$$

By (4.4), $p=3c+(r+1)d+\frac{d}{2}>\frac{d}{2}$. Thus $\frac{d}{2}<\min\{p,c,d\}$, a contradiction.

Therefore the case when $s=1$ and $1\leq k\leq\frac{r-3}{2}$ does not occur. By **Step 2**, we have

$$a(E,X_r)\geq\frac{3m-1}{4m-1}.$$

□

5. Proof of the main theorem

In this section, we construct a Kollár component over $X_r\ni 0$ attaining the lower bound in Proposition 4.1 (Construction-Proposition 5.1), and prove Theorem 1.2.

The weighted blow-up in Construction-Proposition 5.1 attains the lower bound, since equality holds in Lemma 3.2(3) for the exceptional divisor F_r and the subgroup $I_{F_r}\subset G_r$ of Definition-Proposition 2.6(3) is μ_r, which has order r rather than $2r$.

Construction-Proposition 5.1. Notation as in Construction 3.1. Let F_r be the exceptional divisor of the weighted blow-up of Y_r with weights $(2mr,2mr,m,2m-1)$ on the coordinates (a,b,u,z), in the sense of [KM98, Definition 4.56]. Then the following hold.

(1) F_r is a G_r-invariant Kollár component over $Y_r\ni 0$.
(2) We have
$$\mathrm{mult}_{F_r}\mathrm{div}(a)=\mathrm{mult}_{F_r}\mathrm{div}(b)=2mr,\quad \mathrm{mult}_{F_r}\mathrm{div}(u)=m,\quad \mathrm{mult}_{F_r}\mathrm{div}(z)=2m-1,$$
and $a(F_r,Y_r)=3m-1$.
(3) $I_{F_r}=\mu_r$, where $I_{F_r}\subset G_r$ is the subgroup in Definition-Proposition 2.6(3) for the quotient morphism q_r.
(4) The Kollár component E_r over $X_r\ni 0$ induced by F_r (Definition-Proposition 2.6(2)) satisfies $a(E_r,X_r)=\frac{3m-1}{4m-1}$.

Proof. Let $w:=(2mr,2mr,m,2m-1)$, let $w(\psi)$ be the w-weight of a polynomial $\psi\in\mathbb{C}[a,b,u,z]$, and let $\mathfrak{m}_w(n)\subset\mathbb{C}[a,b,u,z]$ be the ideal of polynomials of w-weight at least n, as in [KM98, Definition 4.56].

(1) By Lemma 3.3(1), $Y_r\ni 0$ is a klt germ. Since $r=4m-1$, we have

$$w(ab)=w(u^{4r})=4mr,\quad w(u^6z^N)=6m+(2r+3)(2m-1)=4mr+12m-2r-3=4mr+r. \tag{5.1}$$

Thus the lowest weight part of $ab-u^{4r}-u^6z^N$ is $ab-u^{4r}$. Since u^{4r} has no zero on $\mathbb{A}^2_{u,z}\backslash(u=0)$, by Lemma 3.2(1) applied to $\phi=u^{4r}$, $(ab=u^{4r})$ is a normal variety with canonical singularities. In particular, $ab-u^{4r}$ is irreducible. Thus the hypersurface defined by the lowest weight part

of the equation of Y_r has only canonical singularities at the origin, which is the assumption of [Zhu24a, Lemma 2.5]. Hence F_r is a Kollár component over $Y_r \ni 0$ by [Zhu24a, Lemma 2.5].

The formulas for τ and μ_r in Construction 3.1 define linear automorphisms of $\mathbb{A}^4_{a,b,u,z}$. Since $w(a) = w(b)$, they send any monomial M to a non-zero multiple of a monomial of w-weight $w(M)$, so they preserve $\mathfrak{m}_w(n)$ for any $n \geq 0$. Thus any $\sigma \in G_r$ lifts to an automorphism of the weighted blow-up of $\mathbb{A}^4_{a,b,u,z}$ which preserves the birational transform of Y_r and maps F_r onto F_r. Hence

$$\operatorname{mult}_{F_r} \operatorname{div}(\sigma^*\psi) = \operatorname{mult}_{F_r} \operatorname{div}(\psi)$$

for any $\sigma \in G_r$ and any $\psi \in \mathcal{O}_{Y_r,0}\backslash\{0\}$, that is, F_r is G_r-invariant.

(2) Let μ_{2m-1} act on $\mathbb{A}^4_{s,a_1,b_1,u_1}$ by

$$\epsilon \cdot (s, a_1, b_1, u_1) = (\epsilon s, \epsilon^{-2mr} a_1, \epsilon^{-2mr} b_1, \epsilon^{-m} u_1).$$

By [KM98, Definition 4.56], the weighted blow-up of $\mathbb{A}^4_{a,b,u,z}$ with weights w is $\operatorname{Proj} \bigoplus_{n\geq 0} \mathfrak{m}_w(n)$. The degree-zero part of the homogeneous localization of $\bigoplus_{n\geq 0} \mathfrak{m}_w(n)$ at $z \in \mathfrak{m}_w(2m-1)$ consists of the fractions ψ/z^k with $\psi \in \mathfrak{m}_w(k(2m-1))$, and $\psi/z^k \to \psi z^{-k}$ is an isomorphism from it onto the ring of μ_{2m-1}-invariants of $\mathbb{C}[s, a_1, b_1, u_1]$ under the substitution

$$a = s^{2mr} a_1, \quad b = s^{2mr} b_1, \quad u = s^m u_1, \quad z = s^{2m-1}. \tag{5.2}$$

Thus the open subset of the weighted blow-up of $\mathbb{A}^4_{a,b,u,z}$ where z does not vanish is $\mathbb{A}^4_{s,a_1,b_1,u_1}/\mu_{2m-1}$, with the morphism to $\mathbb{A}^4_{a,b,u,z}$ given by (5.2). By (5.1) and (5.2),

$$ab - u^{4r} - u^6 z^N = s^{4mr}(a_1 b_1 - u_1^{4r} - s^r u_1^6).$$

Let

$$V := (a_1 b_1 = u_1^{4r} + s^r u_1^6) \subset \mathbb{A}^4_{s,a_1,b_1,u_1}.$$

The polynomial $a_1 b_1 - u_1^{4r} - s^r u_1^6$ has degree one in a_1 and b_1 does not divide $u_1^{4r} + s^r u_1^6$, so V is irreducible. Since $V \not\subset (s = 0)$ and the preimage of $0 \in Y_r$ is contained in $(s = 0)$, the birational transform of Y_r in $\mathbb{A}^4_{s,a_1,b_1,u_1}/\mu_{2m-1}$ is V/μ_{2m-1}, and $F_r \cap (V/\mu_{2m-1})$ is the image of $(s = 0) \cap V$.

Let $V^\circ := V \cap (a_1 \neq 0)$. Then V° is the graph of $b_1 = (u_1^{4r} + s^r u_1^6)/a_1$ over $(a_1 \neq 0) \subset \mathbb{A}^3_{s,a_1,u_1}$. In particular, V° is smooth, $(s = 0) \cap V^\circ$ is irreducible, and $a_1^{-1} da_1 \wedge du_1 \wedge ds$ is a nowhere vanishing 3-form on V°. The point $p := (0, 1, 1, 1)$ belongs to $(s = 0) \cap V^\circ$. If $\epsilon \in \mu_{2m-1}$ fixes p, then $\epsilon^m = 1$, and since m and $2m - 1$ are coprime, $\epsilon = 1$. Thus $V \to V/\mu_{2m-1}$ is étale near p, and the image of $(s = 0) \cap V^\circ$ contains a dense open subset of F_r. Hence for any non-zero regular function ψ on Y_r, $\operatorname{mult}_{F_r} \operatorname{div}(\psi)$ is the largest integer k such that s^k divides the pullback of ψ to V°. Since a_1, b_1, and u_1 are units at p, (5.2) implies

$$\operatorname{mult}_{F_r} \operatorname{div}(a) = \operatorname{mult}_{F_r} \operatorname{div}(b) = 2mr, \quad \operatorname{mult}_{F_r} \operatorname{div}(u) = m, \quad \operatorname{mult}_{F_r} \operatorname{div}(z) = 2m - 1.$$

Let $\Omega := a^{-1} da \wedge du \wedge dz$. By Lemma 3.3(1) and Lemma 3.2(2), Ω generates ω_{Y_r}, so $K_{Y_r} = \operatorname{div}(\Omega) = 0$. By (5.2),

$$a^{-1} da = 2mrs^{-1} ds + a_1^{-1} da_1, \quad du = ms^{m-1} u_1 ds + s^m du_1, \quad dz = (2m-1)s^{2m-2} ds,$$

hence

$$\Omega = (2m-1)s^{3m-2} a_1^{-1} da_1 \wedge du_1 \wedge ds. \tag{5.3}$$

Since $a_1^{-1} da_1 \wedge du_1 \wedge ds$ does not vanish on V° and $V \to V/\mu_{2m-1}$ is étale near p, (5.3) implies $\operatorname{mult}_{F_r} \operatorname{div}(\Omega) = 3m - 2$. Thus

$$a(F_r, Y_r) = 1 + \operatorname{mult}_{F_r} \operatorname{div}(\Omega) = 3m - 1.$$

(3) The group G_r acts faithfully on Y_r and fixes 0. By Lemma 3.3(1) and (4), $Y_r \ni 0$ is a klt germ, q_r is étale in codimension one, and $q_r^{-1}(0) = \{0\}$. Thus the assumptions of Definition-Proposition 2.6 hold for $Y = Y_r$, $y = 0$, $G = G_r$, and $q = q_r$. By (1) and Definition-Proposition 2.6(2), F_r induces a Kollár component E_r over $X_r \ni 0$. Let $e := |I_{F_r}|$.

Let $\zeta \in \mu_r$. Consider the automorphism

$$(s, a_1, b_1, u_1) \to (\zeta^2 s, a_1, b_1, u_1)$$

of $\mathbb{A}^4_{s,a_1,b_1,u_1}$. It commutes with the action of μ_{2m-1}, and it preserves V since $(\zeta^2 s)^r = s^r$. Since $\zeta^r = 1$ and $4m-2 = r-1$, by (5.2) it covers the automorphism

$$(a, b, u, z) \to (\zeta^{4mr} a, \zeta^{4mr} b, \zeta^{2m} u, \zeta^{4m-2} z) = (a, b, \zeta^{2m} u, \zeta^{-1} z)$$

of Y_r, which is the action of ζ in Construction 3.1. Thus the induced automorphism of V/μ_{2m-1} is the lift of ζ, and it restricts to the identity on the image of $(s = 0) \cap V$. Since V/μ_{2m-1} and the associated prime blow-up of F_r are both normal near the generic point of F_r and F_r is a prime divisor on both of them, the birational map between them over Y_r is an isomorphism near the generic point of F_r by [KM98, Remark 2.23], and it commutes with the lifts of ζ. Hence the lift of ζ to the associated prime blow-up of F_r restricts to the identity on a dense open subset of F_r, hence on F_r. Thus $\mu_r \subset I_{F_r}$, and r divides e.

By Lemma 3.3(3), $u^2 z = q_r^* \psi$, where ψ is the regular function on X_r induced by the μ_r-invariant function yz on X'_r. By (2),

$$\operatorname{mult}_{F_r} \operatorname{div}(u^2 z) = 2m + (2m-1) = r.$$

By Definition-Proposition 2.6(3), $r = e \operatorname{mult}_{E_r} \operatorname{div}(\psi)$, so e divides r. Thus $e = r$. Since $\mu_r \subset I_{F_r}$ and $|\mu_r| = r = e$, we have $I_{F_r} = \mu_r$.

(4) By (2), (3), and Definition-Proposition 2.6(3),

$$a(E_r, X_r) = \frac{a(F_r, Y_r)}{e} = \frac{3m-1}{4m-1}.$$

□

Remark 5.2. Notation as in Construction 3.1. We have $\operatorname{mld}(X_r \ni 0) = \frac{2m}{4m-1}$, and $\operatorname{mld}(X_r \ni 0)$ is attained by a prime divisor over $X_r \ni 0$ which is not a Kollár component. In particular, $\operatorname{mld}(X_r \ni 0) < \frac{3m-1}{4m-1} = \operatorname{mld}^{\mathrm{K}}(X_r \ni 0)$, and $\operatorname{mld}(X_{4m-1} \ni 0) = \frac{1}{2} + \frac{1}{2(4m-1)}$ is strictly decreasing in m.

For the lower bound, let E be a prime divisor over X_r with center 0, and let F be a prime divisor over Y_r lying over E, with ramification index e_F. Since q_r is étale in codimension one and $K_{Y_r} = q_r^* K_{X_r}$ (Lemma 3.3(4)(5)), the ramification formula implies $a(F, Y_r) = e_F a(E, X_r)$, and

$$\operatorname{mult}_F \operatorname{div}(q_r^* \phi) = e_F \operatorname{mult}_E \operatorname{div}(\phi)$$

for any non-zero regular function ϕ on a neighborhood of $0 \in X_r$. Let c and d be the orders of u and z along F. Since $q_r^{-1}(0) = \{0\}$, we have $c, d \geq 1$. Since z^r and $u^2 z$ are G_r-invariant, they are pullbacks of regular functions on X_r, so e_F divides rd and $2c + d$. By Lemma 3.3(1) and Lemma 3.2(3),

$$a(E, X_r) \geq \frac{c+d}{e_F} = \frac{2c+d}{2e_F} + \frac{rd}{2re_F} \geq \frac{1}{2} + \frac{1}{2r} = \frac{2m}{4m-1}.$$

For the equality, let F' be the exceptional divisor of the weighted blow-up of Y_r with weights $(2r, 3r, 2m-1, 1)$ on the coordinates (a, b, u, z) [KM98, Definition 4.56]. Since z has weight 1, the open subset of the weighted blow-up of $\mathbb{A}^4_{a,b,u,z}$ where z does not vanish is $\mathbb{A}^4_{s,a_1,b_1,u_1}$, with $a = s^{2r} a_1$, $b = s^{3r} b_1$, $u = s^{2m-1} u_1$, and $z = s$. The monomials ab and $u^6 z^N$ have weight $5r$, since $6(2m-1) + 2r + 3 = 5r$, and u^{4r} has weight $4r(2m-1) > 5r$. Thus the birational transform of Y_r in $\mathbb{A}^4_{s,a_1,b_1,u_1}$ is $(a_1 b_1 = u_1^6 + s^{(8m-9)r} u_1^{4r})$, and F' is the irreducible divisor $(s = 0)$ on it. At a general point of F', a_1, b_1, and u_1 are units and the birational transform is smooth, so the orders of a, b, u, and z along F' are $2r$, $3r$, $2m-1$, and 1. By Lemma 3.3(1) and Lemma 3.2(2), $a^{-1} da \wedge du \wedge dz$ generates ω_{Y_r}, and on the chart it equals $s^{2m-1} a_1^{-1} da_1 \wedge du_1 \wedge ds$, where $a_1^{-1} da_1 \wedge du_1 \wedge ds$ does not vanish at a general point of F'. Hence $a(F', Y_r) = 2m$. Since

$\zeta^{2r} = \zeta^{3r} = \zeta^{4m-1} = 1$ for $\zeta \in \mu_r$, the lift of ζ to the chart is $(s, a_1, b_1, u_1) \to (\zeta^{-1}s, a_1, b_1, u_1)$. Since

$$\operatorname{mult}_{F'} \operatorname{div}(\tau^* a) = \operatorname{mult}_{F'} \operatorname{div}(b) = 3r \neq 2r = \operatorname{mult}_{F'} \operatorname{div}(a),$$

we have $\tau(F') \neq F'$. Since μ_r has index 2 in G_r, the stabilizer of F' in G_r is μ_r, and near a general point of F' the quotient by μ_r is $s \to s^r$, and the double cover $Y_r/\mu_r \to X_r$ is not ramified along the prime divisor over Y_r/μ_r below F', since $\tau(F') \neq F'$. Thus the ramification index of F' over the prime divisor E' over X_r below F' is r, and $a(E', X_r) = \frac{2m}{r} = \frac{2m}{4m-1}$.

Finally, F' and $\tau(F')$ are two distinct prime divisors over Y_r lying over E'. If E' were a Kollár component over $X_r \ni 0$, then, with $q' : Y' \to X'$ as in Definition-Proposition 2.6(2), any prime divisor over Y_r lying over E' would be the prime divisor $q'^{-1}(E')$ on Y', since q' is finite. Thus E' is not a Kollár component.

Proof of Theorem 1.2. Let $m \geq 2$ be an integer and $r := 4m - 1$. By Lemma 3.3(3), X'_r is normal. By Lemma 3.3(5), K_{X_r} is $\mathbb{Q}$-Cartier and $X_r \ni 0$ is klt. Since Y_r is a hypersurface in $\mathbb{A}^4$ and q_r is finite, $\dim X_r = 3$. By Proposition 4.1, $a(E, X_r) \geq \frac{3m-1}{4m-1}$ for any Kollár component E over $X_r \ni 0$, and by Construction-Proposition 5.1(4), equality holds for $E = E_r$. Thus, by Definition 2.3,

$$\operatorname{mld}^{\mathrm{K}}(X_r \ni 0) = a(E_r, X_r) = \frac{3m-1}{4m-1}.$$

Finally,

$$\frac{3(m+1)-1}{4(m+1)-1} - \frac{3m-1}{4m-1} = \frac{(3m+2)(4m-1) - (3m-1)(4m+3)}{(4m+3)(4m-1)} = \frac{1}{(4m+3)(4m-1)} > 0,$$

so the sequence $\{\operatorname{mld}^{\mathrm{K}}(X_{4m-1} \ni 0)\}_{m \geq 2}$ is strictly increasing. □

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